

ELECTRICAL CIRCUITS 1 (DC)

LECTURE 3: RESISTANCE

Reference Books

Introductory Circuit Analysis

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OBJECTIVES

- ❑ Become familiar with the parameters that determine the resistance of an element and be able to calculate the resistance from the given dimensions and material characteristics.
- ❑ Understand the effects of temperature on the resistance of a material and how to calculate the change in resistance with temperature.
- ❑ Become familiar with the broad range of commercially available resistors available today and how to read the value of each from the color code or labeling.

INTRODUCTION

❑ The resistance of any material with a uniform cross-sectional area is determined by the following factors:

- ❑ Material
- ❑ Length
- ❑ Cross-sectional Area
- ❑ Temperature



FIG. 3.1

Resistance symbol and notation.

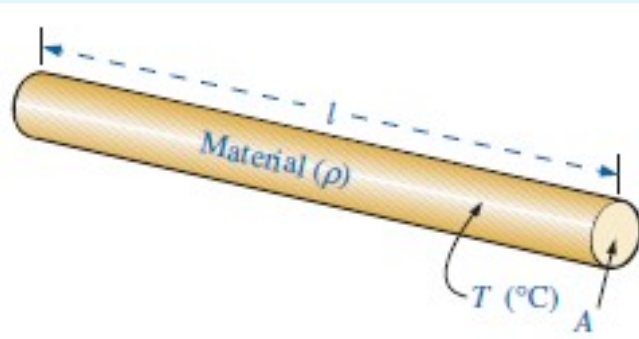


TABLE 3.1

Resistivity (ρ) of various materials.

Material	ρ (CM · Ω /ft)@20°C
Silver	9.9
Copper	10.37
Gold	14.7
Aluminum	17.0
Tungsten	33.0
Nickel	47.0
Iron	74.0
Constantan	295.0
Nichrome	600.0
Calorite	720.0
Carbon	21,000.0

$$R = \rho \frac{l}{A}$$

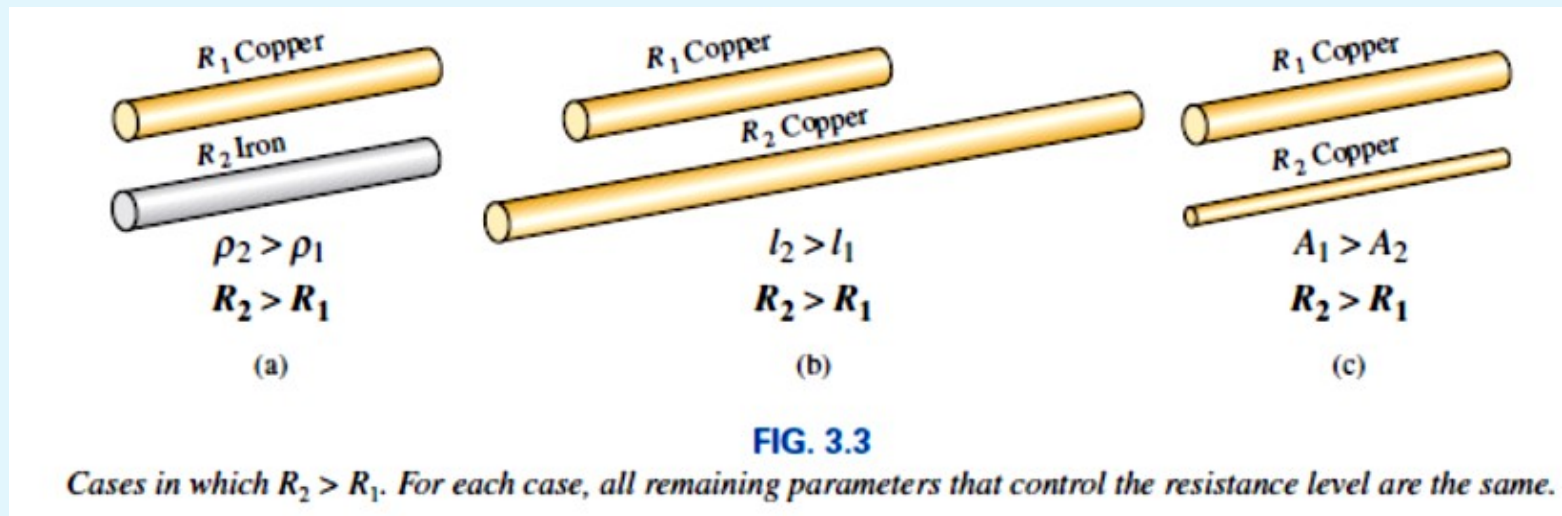
ρ = CM- Ω /ft at $T = 20^\circ\text{C}$

l = feet

A = area in circular mils (CM)

INTRODUCTION

- ❑ The higher the resistivity, the greater the resistance of a conductor.
- ❑ The longer the conductor, the greater the resistance.
- ❑ The greater the area of a conductor, the less the resistance.



TEMPERATURE EFFECTS

- ❑ Temperature has a significant effect on the resistance of conductors, semiconductors and insulators.
- ❑ For good conductors, an increase in temperature will result in an increase in the resistance level. Consequently, conductors have positive temperature coefficients.
- ❑ For semiconductor materials, an increase in temperature will result in a decrease in the resistance level. Consequently, semiconductors have negative temperature coefficients.
- ❑ As with semiconductors, an increase in temperature will result in a decrease in the resistance of an insulator. The result is a negative temperature coefficient.

TEMPERATURE EFFECTS

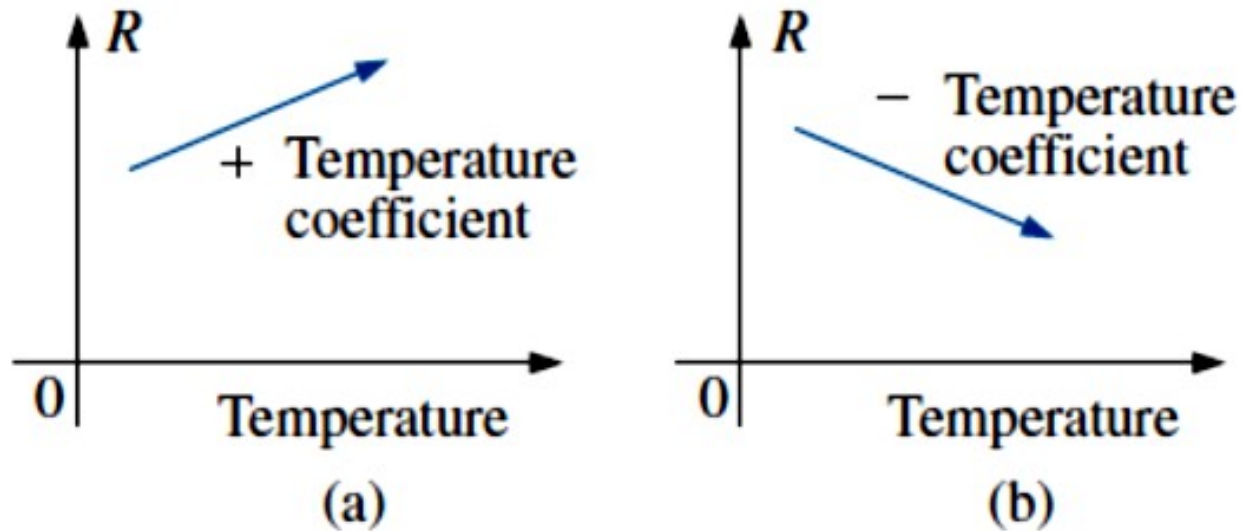


FIG. 3.12

Demonstrating the effect of a positive and a negative temperature coefficient on the resistance of a conductor.

TYPES OF RESISTORS

- Resistors are made in many forms but all belong in either of two groups:
 - Fixed resistors – are made of metal films, high-resistance wire or carbon composition.

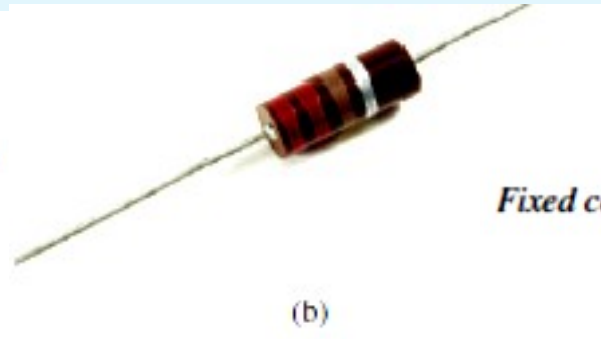
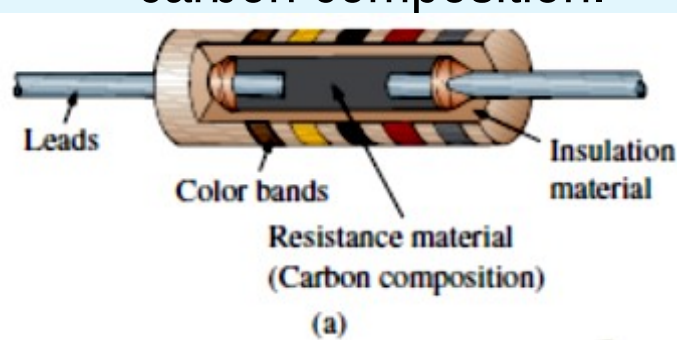


FIG. 3.17
Fixed composition resistors: (a) construction; (b) appearance.

- Variable resistors – have a terminal resistance that can be varied by turning a dial, knob, screw, or anything else appropriate for the application.

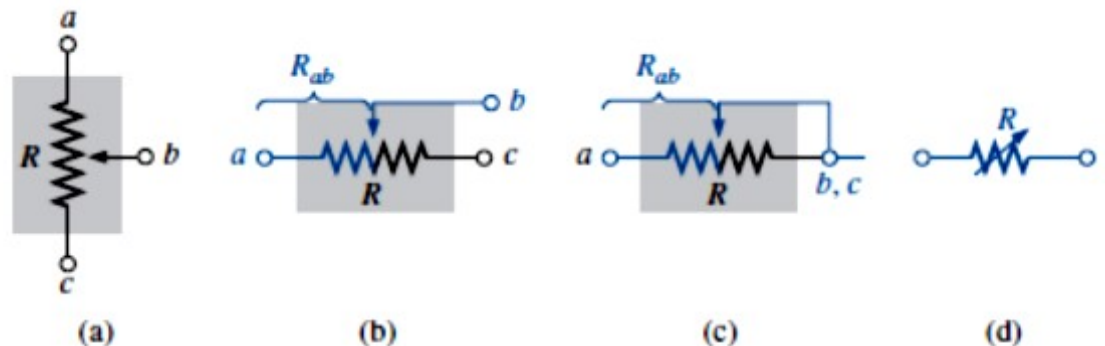
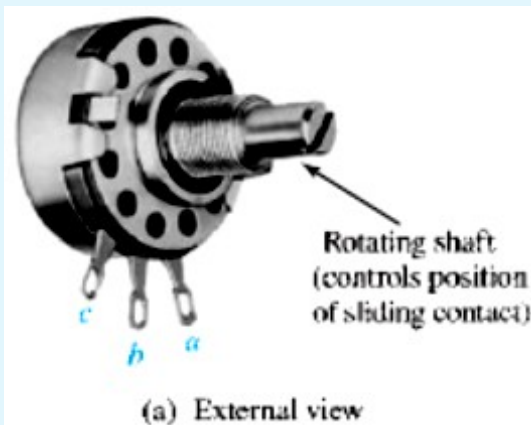


FIG. 3.20
Potentiometer: (a) symbol; (b) and (c) rheostat connections; (d) rheostat symbol.

TYPES OF RESISTORS

- ❑ Variable resistors can have two or three terminals. Most have three.
- ❑ Variable resistors are classified as a rheostat or a potentiometer, depending upon the application.
 - ❑ Rheostat: Two- or three-terminal device used as a variable resistor
 - ❑ Potentiometer: Three-terminal device used for controlling potential levels.

COLOR CODING AND STANDARD RESISTOR VALUES

- ❑ Color coding was developed to identify resistors that were too small for their resistance value to be printed on them.
- ❑ Color bands are always read from the end that has the bands closest to it.
- ❑ 1st and 2nd band represent the first two digits
- ❑ 3rd band determines the power-of-ten multiplier (the number of zeros following the second digit)
- ❑ 4th band is the manufacturer's tolerance (precision of the resistor)

Color	Number		
Black	0		
Brown	1		
Red	2		
Orange	3		
Yellow	4		
Green	5		
Blue	6		
Violet	7		
Gray	8		
White	9		
		±5% (0.1 multiplier if 3rd band)	±10% (0.01 multiplier if 3rd band)

EXAMPLE 3.13

- Find the value of the resistor in Fig. 3.27.



- Reading from the band closest to the left edge, we find that the first two colors of brown and red represent the numbers 1 and 2, respectively. The third band is orange, representing the number 3 for the power of the multiplier as follows:

$$12 \times 10^3 \Omega$$

- resulting in a value of 12 kohm. As indicated above, if 12 kohm is written as 12,000 ohm, the third band reveals the number of zeros that follow the first two digits.

$$5\% \Rightarrow 0.05$$

- Then multiply the resistor value by this decimal number: $0.05(12 \text{ k}\Omega) = 600 \Omega$

$$\text{Maximum} = 12,000 \Omega + 600 \Omega = 12.6 \text{ k}\Omega$$

$$\text{Minimum} = 12,000 \Omega - 600 \Omega = 11.4 \text{ k}\Omega$$

$$\text{Range} = 11.4 \text{ k}\Omega \text{ to } 12.6 \text{ k}\Omega$$

CONDUCTANCE

- ❑ The reciprocal of resistance is conductance (G), measured in siemens (S)

$$G = 1/R \quad (\text{siemens, S})$$

- ❑ A resistance of 1 Mohm is equivalent to a conductance of 10^{-6} S and a resistance of 10 ohm is equivalent to a conductance of 10^{-1} S.

APPLICATIONS

- ❑ Electric baseboard heating element
 - ❑ Heat is generated by passing current through a resistive element.

- ❑ Dimmer controls in an automobile
 - ❑ A two-point rheostat can be used to control light intensity on the dashboard and accessories of an automobile.

- ❑ Strain gauges
 - ❑ Change in the shape of a structure can be detected using strain gauges whose resistance will change with applied stress or flex.